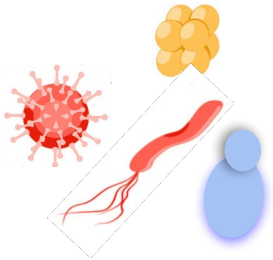
mNGS

metagenomic Next-Generation Sequencing

Key points to learn in this session

- What is mNGS and why is it useful?
- Terminology and types of sequencing
- Use cases
- Other genomics tools in infectious diseases
- Sequencing technologies

Introduction



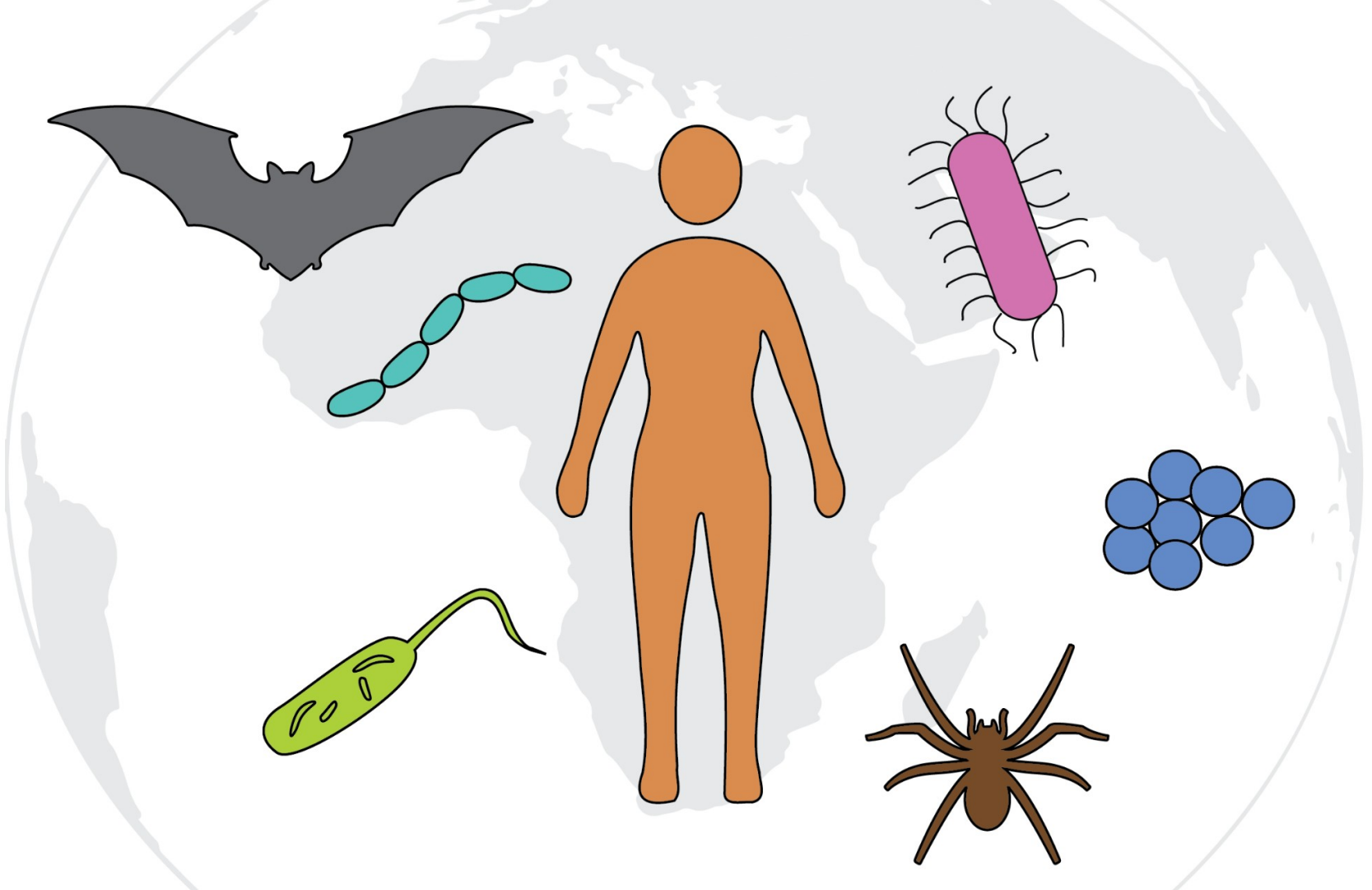
Pathogen landscape

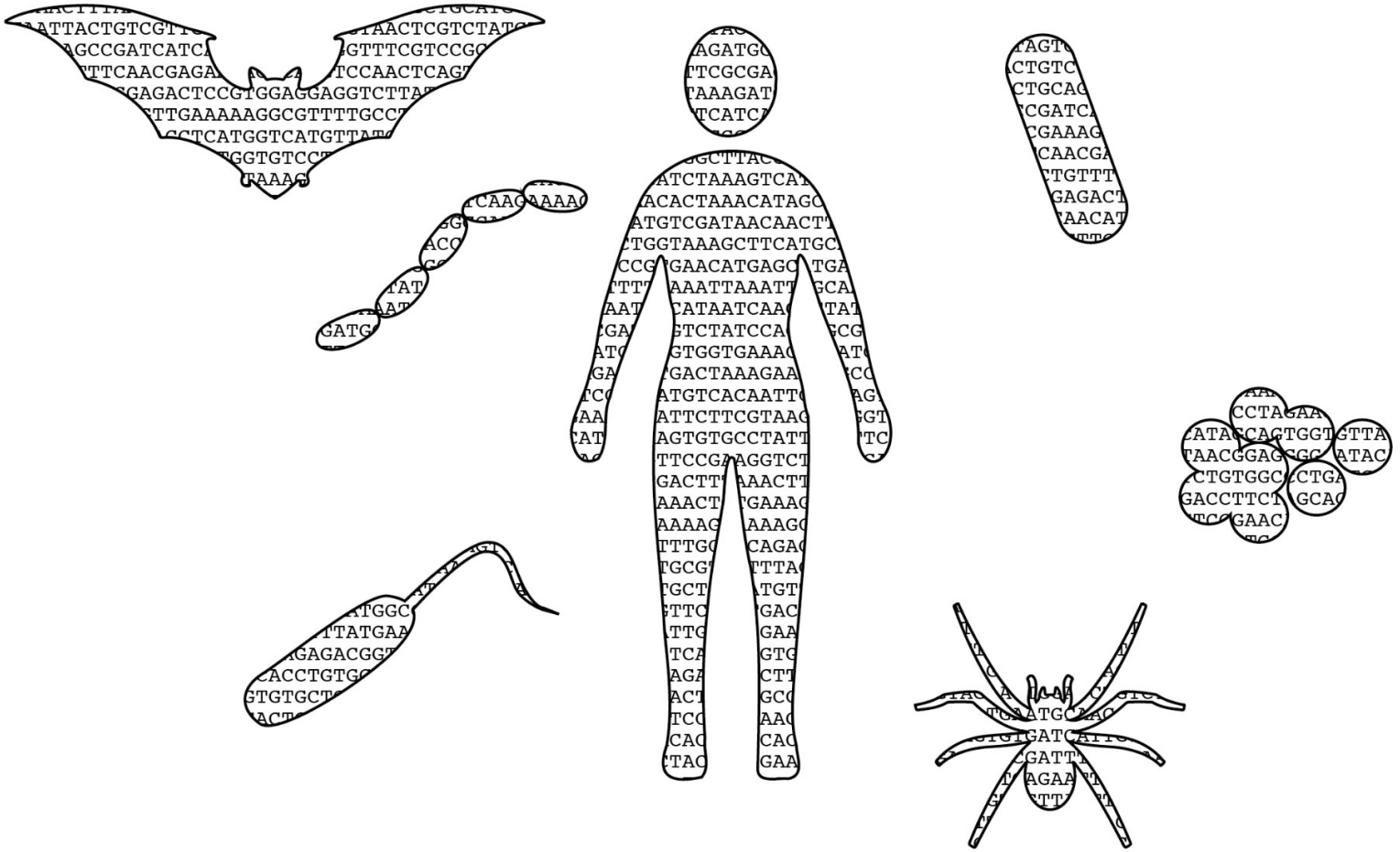


Sample landscape



NGS toolbox for infectious diseases





mNGS



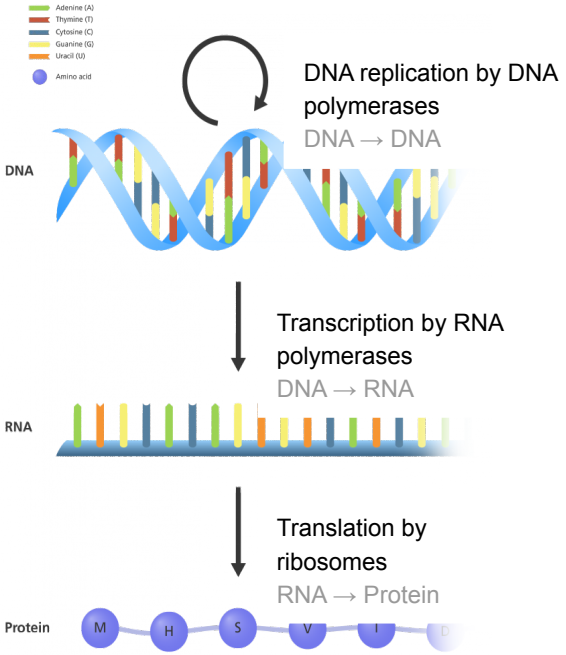


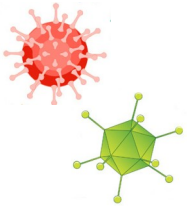
Metagenomic next-generation sequencing (mNGS) is the comprehensive analysis of all of the genetic material present in a sample

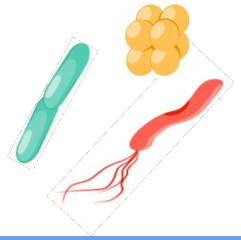
Why is it useful?

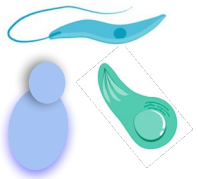
Because it is an **unbiased** approach to detecting and measuring nucleic acids, unlike other methods…

Organisms vary widely in their genetic makeup







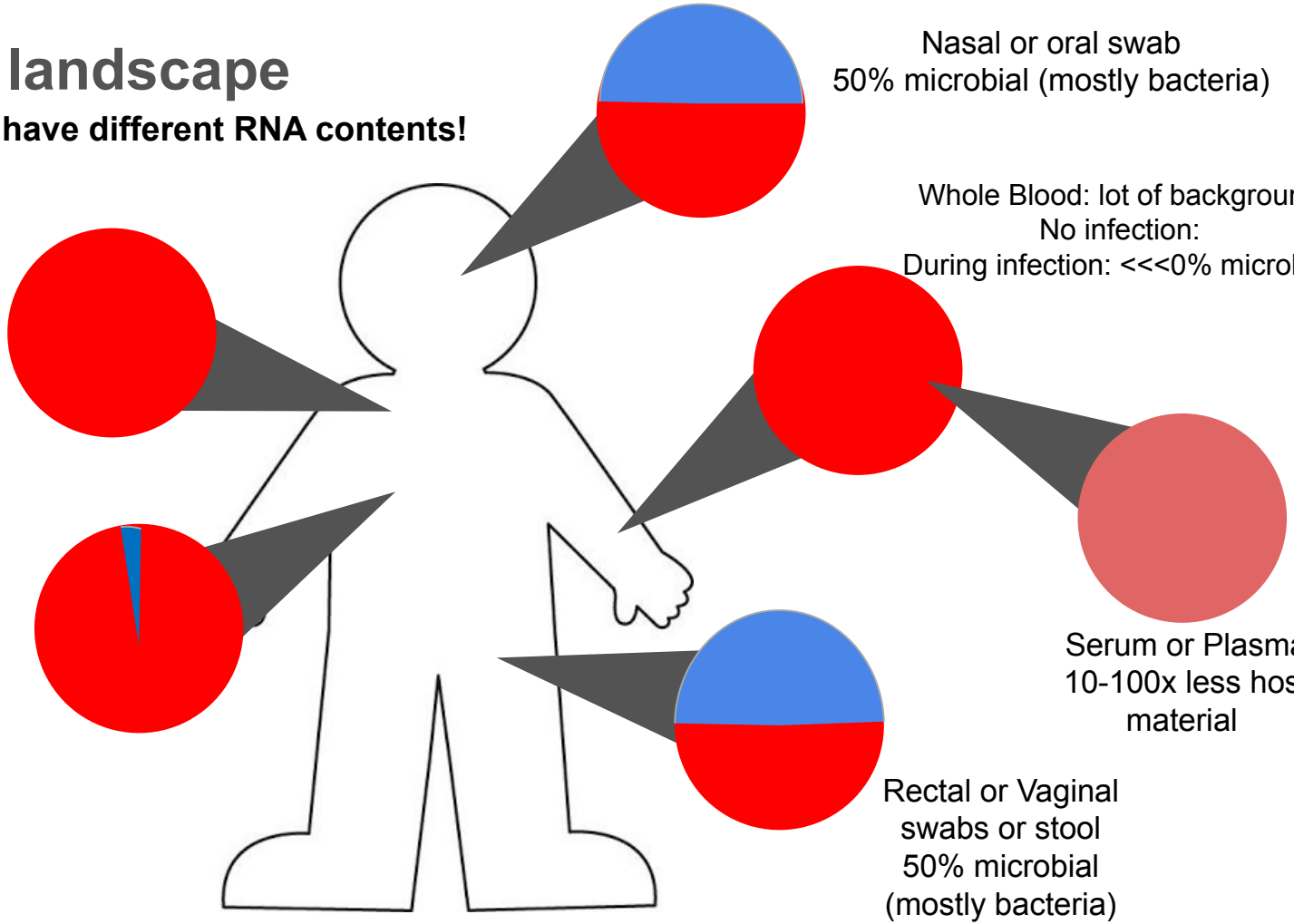


	Viruses	Bacteria	Eukaryotic microbes (yeast, amoebas, worms, etc)
Genome composition	ssRNA, ssDNA, dsDNA, or dsRNA	dsDNA	dsDNA
Ribosomal RNA	None, uses host ribosomes	16S rRNA	18S rRNA
Genome size	2 Kb - 1 Mb	2 Mb - 8 Mb	200 Kb - 100,000 Mb

Adapted from yourgenome.org

Sample type landscape

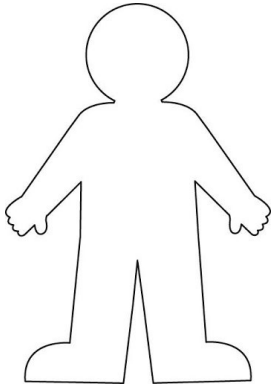
Different sample types have different RNA contents!



Different sample types have different microbial nucleic acid content!



Whole animal/vector:
98% host
1.99% normal flora
0.01% vector borne pathogen



Human samples:
High content: stool, rectal swab
Low content: plasma, CSF



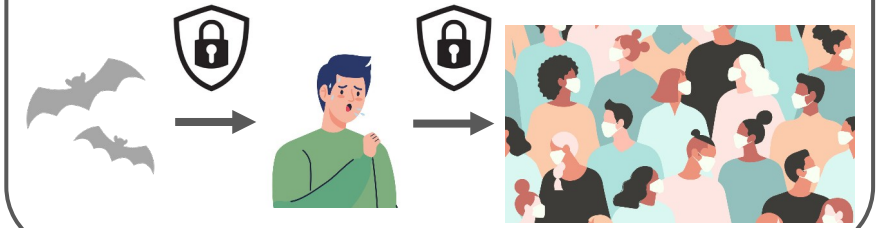
Environmental samples:
High content: wastewater
Low content: aerosol samples, swabs

Use cases for mNGS

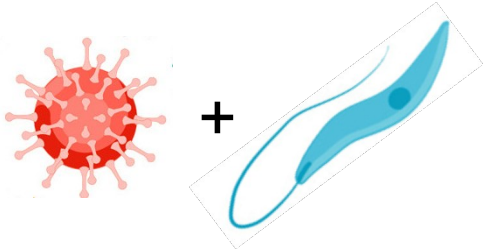
Identify a novel pathogen



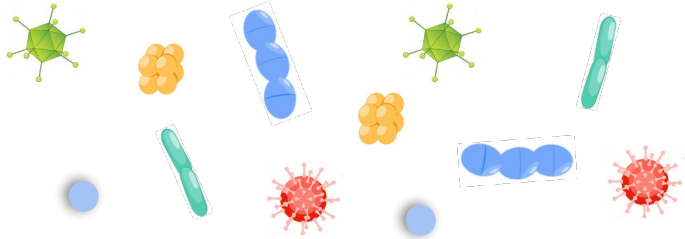
Outbreak detection and prevention



Identify coinfections



Characterize the microbial landscape



Genomics tools in infectious diseases

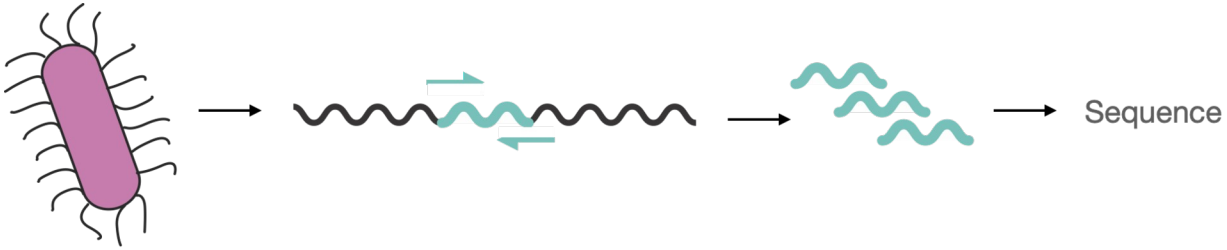
sequencing fragments with Sanger sequencing

WGS

Amplicon sequencing

16S sequencing

Metagenomics NGS



Single sample

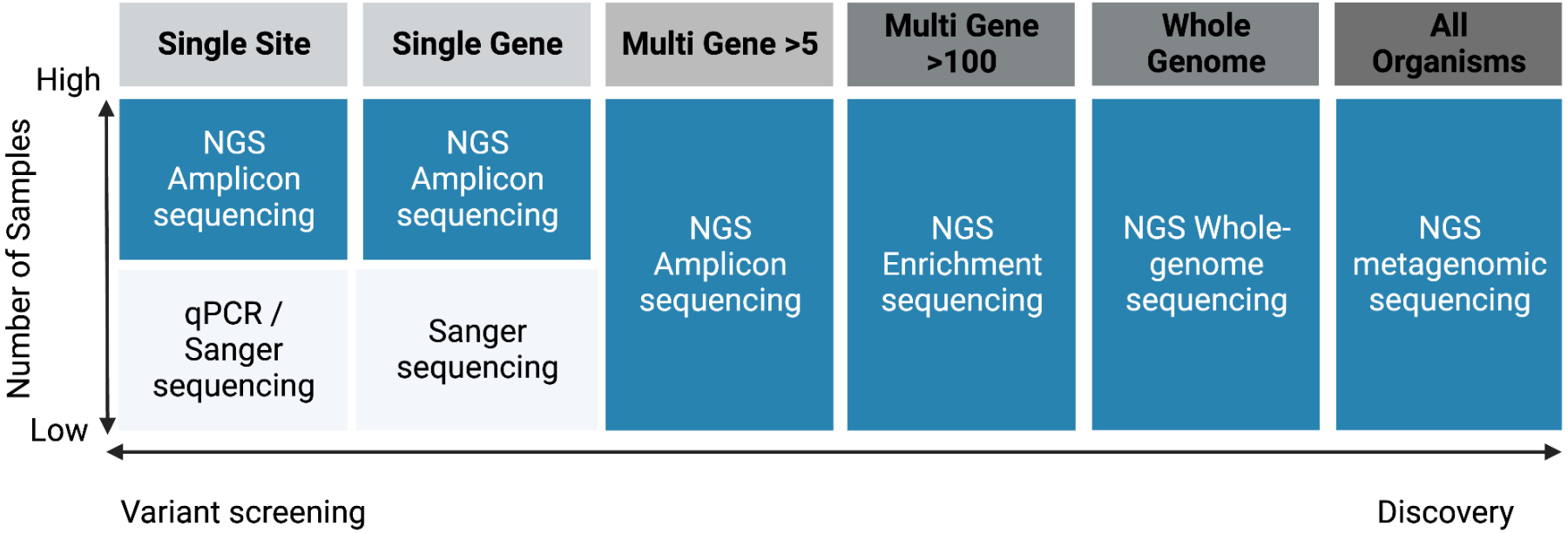
PCR enrichment of a single target

Sanger Sequence (low throughput)

Sanger sequencing

Input	1. One enriched DNA template from a single sample 2. Primers complementary to the ends of the DNA target template
Output	Sequence of one fragment ~500-600bp or one PCR product or plasmid
Strengths	• Fast and cheap • High accuracy
Uses	• Screen for a variant(s) in a specific known genomic region • Often used to confirm cloning products/plasmids • Confirm NGS findings
Limitations	• Short sequence lengths • Low throughput ○ One sample at a time ○ One region/target at a time • Requires large amount of DNA input • Requires knowledge of the target

Sanger sequencing vs Next Generation Sequencing applications



Adapted using BioRender.com from https://www.illumina.com/science/technology/next-generation-sequencing/ngs-vs-sanger-sequencing.html

Whole genome sequencing

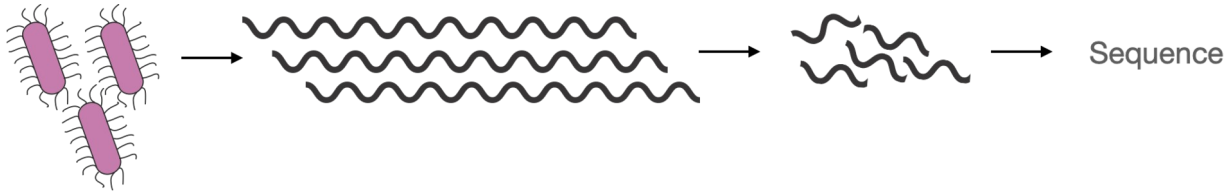
Sanger sequencing

WGS

Amplicon sequencing

16S sequencing

Metagenomics NGS



Single organism

High molecular weight DNA

Fragmentation
(short-read platforms)

Whole genome sequencing

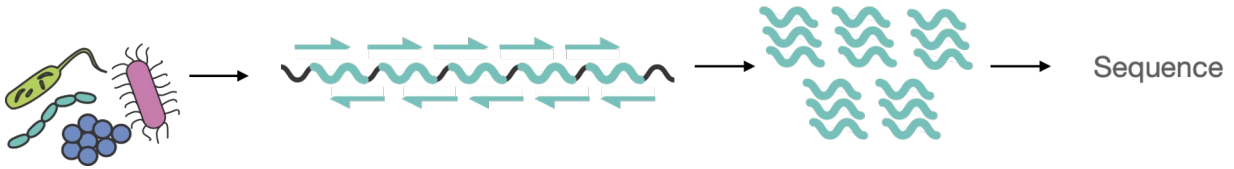
Input	High molecular weight DNA from a cultured isolate
Output	NGS sequencing of the genome of the cultured organism
Strengths	• Does not require specific primer design • Good genome coverage of organisms even with large genomes
Uses	• Obtain whole genome coverage of organisms even with large genomes • Identify or confirm identity of a cultured organism • Identify genomic variants within an organism
Limitations	• Requires a cultured isolate ○ Time-intensive ○ Not all organisms can be cultured; many require special conditions • More hands-on time due to culture requirement

Amplicon sequencing

Sanger sequencing

WGS

Amplicon sequencing



Complex sample

Multiplex PCR enrichment of a multiple targets

(in this case, tiled PCR enrichment of a whole genome of a single organism)

16S sequencing

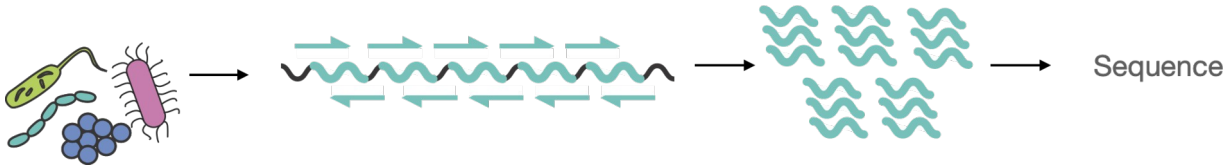
Metagenomics NGS

Amplicon sequencing

Sanger sequencing

WGS

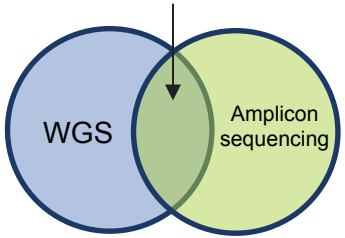
Amplicon sequencing



Complex sample

Multiplex PCR enrichment of a multiple targets

(in this case, tiled PCR enrichment of a whole genome of a single organism)



16S sequencing

Metagenomics NGS

Amplicon sequencing

Input	1. Direct clinical sample or other sample type; does not require a cultured isolate 2. Primers complementary to the target regions / organism of interest
Output	NGS sequence of many PCR products in parallel; genome of organism of interest
Strengths	• Good coverage of whole genome of relatively small genomes • Does not require a culture isolate; can be a mixed/direct sample • The majority of resulting data is of the desired target
Uses	• Genomic epidemiology • Transmission investigation • Identification or monitoring of variants
Limitations	• Requires good primer design and validation • Requires knowledge of the target • Can be difficult for large genomes • Not robust to mutations in primer binding sites

16s sequencing

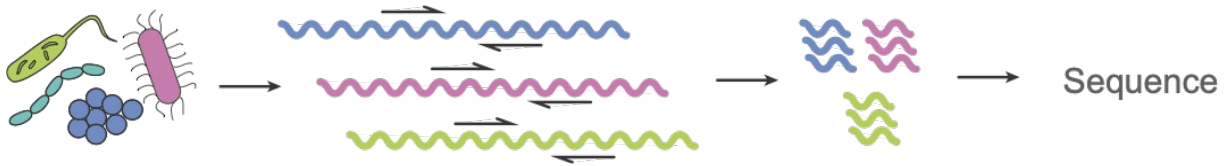
Sanger sequencing

WGS

Amplicon sequencing

16S sequencing

Metagenomics NGS



Complex sample

PCR enrichment of ribosomal gene target (16s)

(this gene is conserved across bacterial species)

16s sequencing

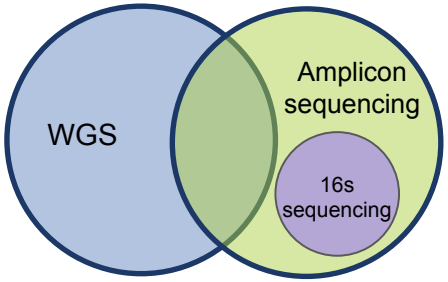
Sanger sequencing

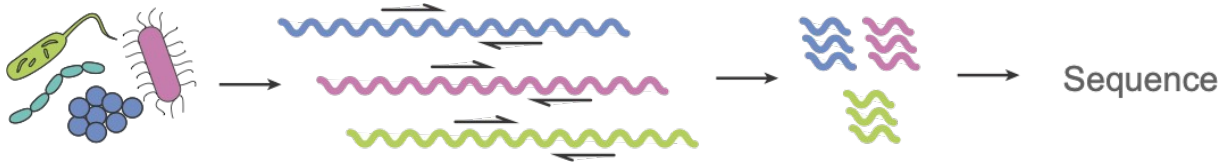
WGS

Amplicon sequencing

16S sequencing

Metagenomics NGS





Complex sample

PCR enrichment of ribosomal gene target (16s)

(this gene is conserved across bacterial species)

16s sequencing

Input	1. Direct clinical sample or other sample type; does not require a cultured isolate 2. Primers complementary to the 16s rRNA gene (V1, V2, or V3 regions)
Output	NGS sequences of amplified 16s rRNA gene target for all but only bacteria in the sample
Strengths	• Good sensitivity in picking up low abundance targets • Does not require a culture isolate; can be a mixed/direct sample • The majority of resulting data is of the desired target
Uses	• Microbiome studies • Phylogeny and taxonomic classification
Limitations	• Can only detect bacteria (organisms that carry 16s rRNA gene) • Can be hard to resolve closely related species • Does not detect antibiotic resistance genes or targets

Metagenomic sequencing

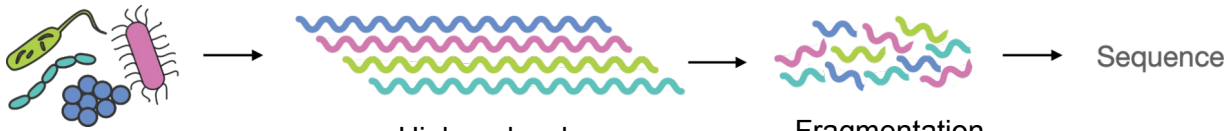
Sanger sequencing

WGS

Amplicon sequencing

16S sequencing

Metagenomics NGS



Complex sample

High molecular weight DNA (ideally)

(if low input, non-targeted amplification will be necessary)

Fragmentation
(short-read platforms)

Metagenomic sequencing

Input	Direct clinical sample or other sample type; does not require a cultured isolate
Output	NGS sequence of all organisms in each sample
Strengths	• Does not require a culture isolate; can be a mixed/direct sample • Unbiased recovery of nucleic acids in a mixed sample • Does not require specific primer design
Uses	• Identification of pathogens with unknown origin • Identification of coinfections • Screening surveillance for multiple organisms with one assay • Outbreak investigation • Pathogen discovery work
Limitations	• Occasionally very low abundance microbes can be missed • Data analysis and interpretation can be difficult • The host is sequenced along with the microbes; can be difficult to remove

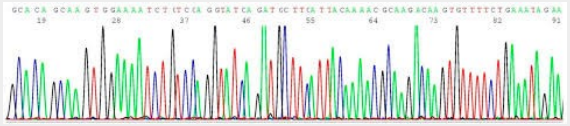
Genomics sequencing toolbox for infectious diseases

	Whole genome & transcriptome sequencing	Amplicon sequencing	16S & 18S rRNA sequencing	Metagenomic (shotgun) sequencing
Starting material	DNA/RNA from (cultured) isolate	DNA/RNA from a sample	DNA/RNA from a sample	DNA/RNA from a sample
Primers used to amplify sequences	Random primers	Organism-specific primers	16S/18S-specific primers	Random primers
What is sequenced?	Organism of interest	Organism of interest	Only rRNA from bacteria, fungi and eukaryotic pathogens in sample	DNA/RNA from all organisms in sample
Resulting data	DNA sequence or gene expression of target organism	RNA/DNA sequence of targeted organism	Only rRNA sequences of bacteria, fungi and eukaryotic pathogens	DNA/RNA sequences from all organisms in sample
Use cases	Fundamental research on cultured organisms	Full genomes from clinical samples for genomic epidemiology	Exploratory studies on environmental samples	Outbreak research, co-infections, microbe discovery

Sequencing technologies

Low throughput, short read sequencing

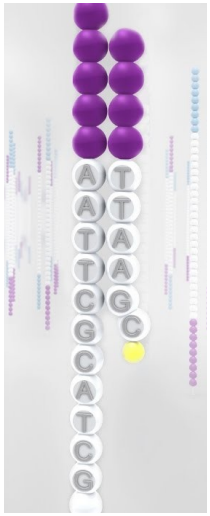
Sanger sequencing



1 x 500bp sequence per well

High throughput, short read sequencing

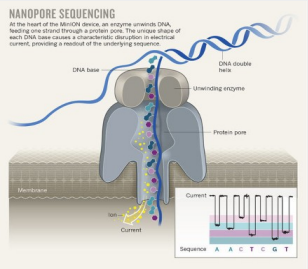
Illumina sequencing



Millions of 300-600bp sequences

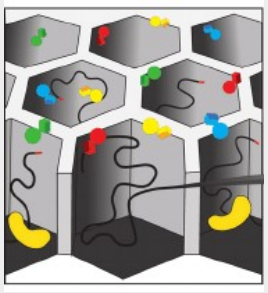
Long read sequencing

Nanopore Sequencing



Thousands to millions of 1-20kb sequences

PacBio sequencing



Thousands of 1 - 30kb sequences

illumina®

High thruput short read sequencing
max. read length 300bp



NovaSeq™ 6000

Cheapest instruments
Low thruput
EOL in 2025
Supported ~2029



NextSeq 550™



MiSeq™



MiniSeq™

iSeq™ 100

High thruput
Still supported
Phased out

XLEAP-SBS (improved chemistry)

MiSeq i100 (Plus option)
5-200 million reads



NextSeq 1000/2000
100 million to 3.6 billion reads



NovaSeqX (Plus option
1.6 – 70 billion reads



Oxford NANOPORE Technologies

High thruput long read sequencing



MinION Mk1D



PromethION Series



GridION

Instruments that will sequence from 1 to 48 flow cells.
The MinION instruments require an external computer

Other sequencing platforms

Long read

PacBio



Short read

ion torrent
by Thermo Fisher Scientific

Element Biosciences

华大基因
BGI

Comparing short and long read sequencing platforms

Feature	*Short-Read Sequencing (e.g. Illumina)*	*Long-Read Sequencing (e.g. Oxford Nanopore)*
Read Length	50-300 bp	100 bp - 100,000 bp
Accuracy	High (99.9% with Illumina)	Lower (~90-99%) but improving
Cost per run	~$800	~$900
Instrument Cost	~$20,000	~$3,000
Throughput (claimed)	1.2 Gb	20 Gb
Throughput (observed)	1.2 to 2 Gb	1-6 Gb
Run Time	Hours to days	Minutes to days (potentially real-time)

Considerations for choosing a sequencer platform

- Available resources
  - Procurement
  - Lead time
  - Frequency of use
  - Expertise and comfort
  - Downstream integration
  - Support
- Objective
  - Identification
  - Variant quality in complex genomes
  - Turnaround time
- Sample
  - Input
  - Quality

mNGS pipeline

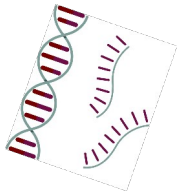
Study design



- Type(s)
- Controls
- Storage
- Lab setup

xNA extraction

(1-2 hrs)



- Type(s)
- Controls
- QC

Library prep

(1-2 days)



- Host removal
- Fragmentation
- dsDNA generation
- End-repair
- Adapter ligation
- Barcoding & PCR
- Clean-up
- Pooling
- QC

Sequencing

(0.5-3 days)



- Machine
- Read length
- Loading conc.

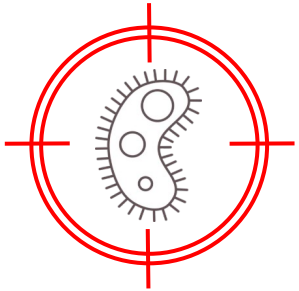
Data analysis

(0.5-4 hrs)



- QC data
- Host removal
- Alignment
- Blast
- Ranking of hits

Pathogen ID



Next: 02_mNGS study design